File Integrity Monitoring in Wazuh

By Varun Wilfred

Introduction

Attackers don't always just break into systems — they change them. For example, ransomware encrypts files, attackers modify logs, and other malicious actions alter system data. Detecting these changes quickly is critical.
File Integrity Monitoring (FIM) is a security practice that tracks file and directory modifications in real time and generates alerts when suspicious activity is detected.
FIM is also a key requirement for many regulatory compliance standards, including PCI-DSS, HIPAA, and ISO 27001.
This guide demonstrates how to configure Wazuh's FIM feature on a Windows endpoint. It assumes Wazuh is already installed and that the Wazuh agent is deployed on the target machine.

2 Configuring Wazuh FIM on Windows

This guide assumes that Wazuh is already installed and the agent is deployed on a Windows machine. To monitor specific files or folders with Wazuh's FIM feature:

1. Open the Wazuh agent configuration file on the endpoint:

```
/var/ossec/etc/ossec.conf
```

2. Within the <syscheck> block, add the directory you want to monitor.
   Example: monitoring a folder on the Windows desktop.

```
<directories check_all="yes" report_changes="yes" realtime="yes">
C:\Users\dell\OneDrive\Bureau
</directories>
```

3. (Optional) Control subdirectory monitoring with the recursion level attribute:

   - 0 – monitor only the specified folder
   - 1 – monitor the folder and immediate subfolders
   - 2 – monitor the folder and all nested subfolders up to two levels deep

4. Save the file and restart the Wazuh agent:

```
sudo systemctl restart wazuh-agent
```

3 Validation: Monitoring in Action

After configuration, Wazuh will begin tracking file activity in the specified path. As shown in Figure 1 and Figure 2, the test file file.docx was:

- Created
- Modified several times
- Deleted

Each of these actions was detected and reported by Wazuh, triggering alerts with appropriate rule levels.

```
<!-- File integrity monitoring -->
<syscheck>

  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>

  <!-- Default files to be monitored. -->
  <directories check_all="yes" report_changes="yes" realtime="yes">C:\Users\dell\OneDrive\Bureau</directories>
  <directories recursion_level="0" restrict="regedit.exe$|system.ini$|win.ini$">%WINDIR%</directories>

  <directories recursion_level="0" restrict="at.exe$|attrib.exe$|cacls.exe$|cmd.exe$|eventcreate.exe$|ftp.exe$|lsass.exe$|net.
  <directories recursion_level="0">%WINDIR%\SysNative\drivers\etc</directories>
  <directories recursion_level="0" restrict="WMIC.exe$">%WINDIR%\SysNative\wbem</directories>
  <directories recursion_level="0" restrict="powershell.exe$">%WINDIR%\SysNative\WindowsPowerShell\v1.0</directories>
  <directories recursion_level="0" restrict="winrm.vbs$">%WINDIR%\SysNative</directories>
```

1: Configuration to monitor Desktop folder in ossec.conf

1,521 hits

June 25, 2026 @ 16:32:44.272 - June 26, 2026 @ 16:32:44.273

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timestamp	agent.name	syscheck.path	syscheck.event	rule.des...	rule.lev
July 15, 2026 @ 16:32:30.053	soc-user	c:\users\dell\onedrive\bureau\test file.docx	deleted	File deleted.	7
July 15, 2026 @ 16:31:36.481	soc-user	c:\users\dell\onedrive\bureau\test file.docx	modified	Integrity ch...	7
July 15, 2026 @ 16:31:36.426	soc-user	c:\users\dell\onedrive\bureau\test file.docx	modified	Integrity ch...	7
July 15, 2026 @ 16:30:30.325	soc-user	c:\users\dell\onedrive\bureau\test file.docx	added	File added ...	5

2: Wazuh alerts showing file creation, modification, and deletion